Python

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Google colab

* Programming :-
programming means giving instructions to a computer to perform a task.

* Programming Language :-
A programming language is a formal language used to communicate instructions to a computer.
ex:- Python, Javam, C, C++, Javascript, C#, Php, .net, ruby, Rust, Kotlin, SQL.

* A - Low level language :-
- These are closer to machine hardware.
- Assembly language.

* B - High level language :-
- easier humans to read and write
- Select * from HR;

* Compiled vs Interpreted Language :-
- C, C++, Rust - Compiled
- Code --- Complier ---- enecute ---- run

- Interpreted language enecutes the code through a runtime rather than a complen. ex- MySQL.

* Syntax -
Syntax means the rule for writing valid code
- If you don't follow the rule - Syntax error

* IDE - Integrated Development Environment

An IDE is a software - that provides tools for writing, running and, debbing and managing the program
ex:-
- WorkBench
- VS Code - visual studio code
- Jupyter Notebook
- Spyder
- Online - Google Colab .py - Python file -IDE .pynb - Python Notebook, .ipynb - Internet python Notebook, .pbix - powerbi file, excel - excel, SQL - mysql, .csv - comma separated visuds, .tsv - tab seperate values.

* Python -
python is a High level general purpose programming language webpage, website, games, AI, APP,
- Easy to read and write.

run → shift + Entre

* Variable - It is a container

- variable name is anything
- not starting with numbers
- variable ~~number~~ name not contains space
- rather than use _

ex:- tushar = 10 , tushar1 = 10 , ~~1tushar = 10~~
1tushar = 10 --- It's wrong
tushar Atkare = 10 --- It's wrong
tushar _Atkare = 10
tushar - Atkare = 10 <-- it's wrong.

- Python is a case sensitive language.
sudhiksha , SUDIKSHA , SuDhiksha , SudhiksHA
- The Shudiksha is not defined.

* Data Types :-
int - 10, 20, 30 , a=10.
float - 10.5 , 9.6
Complex - 2+3j
String - str - " Tushar "
boolien - bool - True/False (True/false is wrong)
list - [10, 20, 30, 40, 50, 60]
set - {10, 20, 30, 50, 60}
tuple - (10, 20, 30, 60)
dictionary - {key : value}
None Type - None

Date.

* Comments - # Single line Comments ;
"" this is multi line comments ""

* Operators - symbols for Calculations ;

+ , - , * , /

// - Floor division % - modular operation (for reminder) ** - power

* Comparision Operator :

< , > , >= , <= , == (equal)

* logical :

Not , And , OR

* Keyword - A Keyword is a reserve word in python with predefine meaning :-

True, False, If , Else, Elif , while, for, def, return, pass, continue, from, import, None, and, or, not, is, in

* Function :-
A reusable block ~~for~~ of code

SUM() , MIN() , PRINT()
- Predefine function - build in function
- User define function - (Sangeet)
- function syntax is ---- function_name() = MIN()

* Method :-
It is a function associated with an object.

name = "tushar"
name.upper() → method ---- Syntax - object.function_name()

* Module :-
- A module is a python file containing code such as function, methods, variable, objects that can be import in another program
- every module need to import first - before using

```
import math
math.sqrt(25)
```